

BRONCHIAL ASTHMA IN ADULTS

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INTRODUCTION

Importance

Asthma is a chronic reversible airway disorder that is common in people of all ages. It can be severe and may be fatal. Asthma may present with cough, wheezing, and unexplained dyspnoea and chest tightness. Symptoms are often transient, may be persistent and tend to be worse at night or in the early mornings. Asthma symptoms may be precipitated or aggravated by upper respiratory tract infections, cigarette smoke, environmental haze, exercise, drugs (e.g. aspirin, NSAIDs, β -blockers, ACE inhibitors), pets and occupational exposure to triggers.

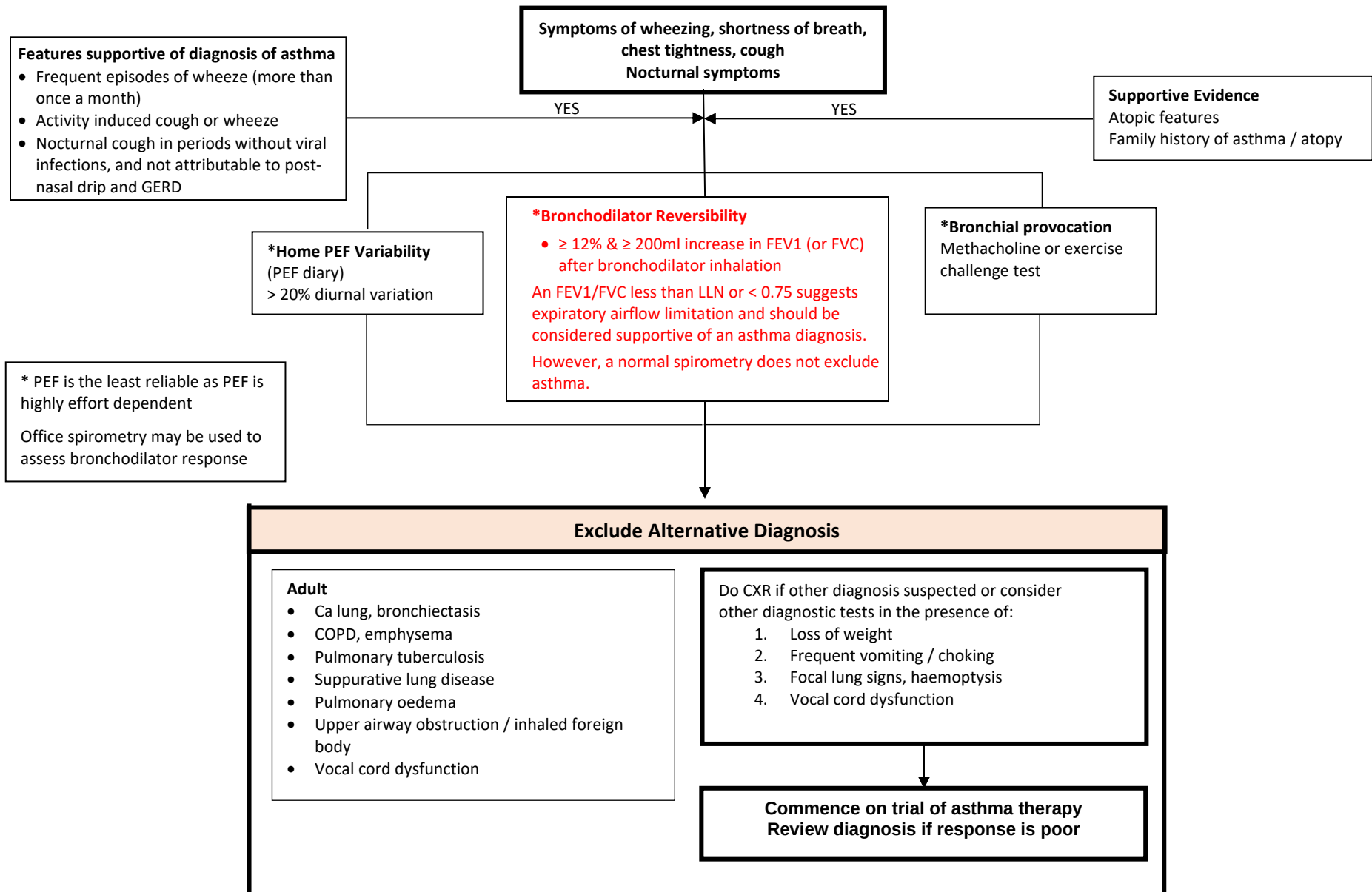
A diagnosis of asthma is based on clinical presentation of characteristic symptoms and where possible, documentation of variable expiratory airflow limitation. Initiation of inhaled corticosteroids should not be delayed as these tests can be normal in mild or well controlled asthma.

Epidemiology

Asthma is one of the most common chronic respiratory conditions seen in primary care in Singapore. Around 5% of residents in Singapore aged 18 to 69 years have asthma. About 1 in 3 patients with asthma aged 12 years and older in Singapore report exacerbations in the past year, and about 1 in 2 have missed work or school due to asthma in the past year. The impact of asthma locally is also reflected in hospital admissions, with Singapore's asthma hospital admission rates being higher than countries in the Organisation for Economic Co-operation and Development (OECD).

Risk of exacerbations and other poor asthma outcomes, such as hospital admissions, can be reduced with preventer (or controller) medications, particularly inhaled corticosteroid (ICS)—the mainstay of long-term asthma management. Despite wide availability of ICS, use of preventers in Singapore is the lowest among eight countries in the Asia-Pacific region, with only 1 in 4 patients with asthma aged 12 years and older using a preventer in the past month. Locally, one third of patients with a severe asthma exacerbation requiring mechanical ventilation or intensive care unit (ICU) admission were not on ICS prior to the exacerbation. To reduce the impact of asthma in Singapore, more optimal ICS use as part of long-term management is needed.

PRESENTATION AND DIAGNOSIS



Other Modes of Presentation

1. Cough variant asthma without wheezing
2. Adult onset asthma: consider referral if patients do not respond well to treatment to exclude eosinophilic granulomatosis with polyangiitis (EGPA), chronic rhinosinusitis and nasal polyps, allergic bronchopulmonary aspergillosis (ABPA). Obesity may also be associated with a higher risk of developing adult-onset asthma.
3. 'First acute wheeze' – Exclude infections, foreign body aspiration, endobronchial lesions (unilateral wheezing)
4. In cigarette smokers, consider COPD with asthma
5. Exercise-induced bronchoconstriction
6. Asthma in pregnancy (1/3 of patients have deterioration of asthma during pregnancy due to hormonal changes)

Investigations

Investigations are usually not necessary except in severe or atypical cases, and in patients who do not respond to therapy. Some investigations which may be considered are:

1. Chest X-ray

To exclude foreign body or chronic chest infection (e.g. pulmonary TB for chronic cough) or to exclude complications in severe acute episodes.

2. Pulmonary Function Tests (Peak expiratory flow rate (PEFR) / Spirometry)

The demonstration of diurnal variation of PEFR $\geq 20\%$ or a positive bronchodilator response on spirometry (post bronchodilator increase in FEV1 by $\geq 12\%$ and ≥ 200 ml) can support a diagnosis of asthma.

Note: Do not delay initiation of ICS treatment if the clinical suspicion for asthma is high or if asthma is poorly controlled. Tests are used to support the diagnosis but can be normal in well controlled asthma and if patient is currently on LAMA/LABA. BDR reversibility can be present in COPD and is not specific for asthma.

3. Allergy Tests

Atopic status can be identified by skin prick testing or by measuring the level of specific immunoglobulin E (sIgE) in serum especially in patients with no clinical signs or family history of atopy, however, the presence of a positive skin prick test or positive sIgE, does not mean that the allergen is causing the symptoms - the relevance of allergen exposure and its relation to symptoms must be confirmed by the patient's history.

Other allergy tests such as antigen specific IgG, IgG4, intradermal skin tests are not useful. Food allergy testing is also not useful for evaluation of asthma per se.

4. Airway Challenge Tests - Using Exercise or with Inhalation of Methacholine or Histamine

Methacholine challenge is a sensitive test to exclude asthma. Exercise challenge is reserved for evaluation of exercise-induced asthma.

5. Fractional concentration of exhaled nitric oxide (FeNO) has not been established as useful for ruling in or ruling out asthma, as it can be elevated in some non-asthma conditions such as eosinophilic bronchitis, atopy, allergic rhinitis and eczema. It may also be not elevated in some asthma phenotypes such as neutrophilic asthma, and lower in smokers and in early stages of allergic responses.

6. Other Tests to Exclude Other Medical Conditions

- CT thorax of the lung: to exclude emphysema, bronchiectasis, lung cancer, tracheobronchomalacia
- Induced Sputum for AFB smear and culture - to exclude Mycobacteria infection
- Nasoendoscopy and/or CT scan of the sinuses to exclude chronic rhinosinusitis with / without nasal polyps that can be associated with adult onset severe asthma.
- OGD, pH manometry to exclude GERD that can mimic asthma
- Bronchoscopy to exclude endobronchial lesions
- Immunological investigations, e.g. HIV, Serum Immunoglobulin titres

MANAGEMENT

Asthma Control Goals and Outcomes

- No limitation of daily activities, including exercise
- No (twice or less/week) daytime symptoms
- No nocturnal symptoms or awakening because of asthma
- No (twice or less/week) need for reliever treatment
- No exacerbations
- Normal or near-normal lung function results

Components of Asthma Management

1. Good doctor-patient relationship
2. Identification and reduction of exposure to risk factors such as dust mites, moulds, pets, pollen and cigarette smoke
3. Assessment, treatment and monitoring of Asthma
4. Management of asthma exacerbations
5. Patient education including Written Asthma Action Plan

Follow-Up Interval

1. Asthma control review
The interval of reassessment can vary from once in 2 weeks (for a patient with poor asthma control undergoing adjustment in medication) to once in 6 months for a patient with very well controlled condition.
2. Annual review of asthma action plan
3. Annual smoking assessment

Monitoring of Adults with Asthma in Primary Care

Factors that should be monitored and recorded include:

- Asthma control using tools such as GINA assessment of Asthma control, Asthma symptom score, Asthma Control Test (ACT)[®]
- Asthma risks:
 - High risk for severe attacks: ≥ 3 attacks or prednisolone bursts in last 12 months, ≥ 3 canisters of SABA used in last 12 months
 - High risk for life-threatening attack: previous ICU admission / intubation for status asthmaticus, ≥ 1 canisters of SABA every month,
 - Other risk factors: chronic airflow limitation (baseline $FEV_1 < 50\%$), persistent poor ICS adherence and smoking (or exposure to cigarette smoke), psychosocial factors.
- Adherence to asthma medication
- Inhaler technique
- Aerochamber / spacer care (to clean every month and to change every 6 to 12 months)
- Check and address causes of poor asthma control
 - Refer asthma nurse to cross-check adherence and technique (50% of patients are not adherent)
 - Check for triggers (aeroallergens, irritants, haze, cigarette smoker)
 - Check for drugs that can aggravate asthma (e.g. aspirin, NSAIDS, non cardioselective β -blockers)
 - Confirm diagnosis of asthma - CXR and spirometry
- Understand use of a self-management plan / written personalised asthma action plan to prevent deterioration during asthma attacks

GINA Assessment of Asthma Control

A. Asthma symptom control		Level of asthma symptom control		
In the past 4 weeks, has the patient had:		Well controlled	Partly controlled	Uncontrolled
• Daytime asthma symptoms more than twice/week?	Yes ☐ No ☐	None of these	1–2 of these	3–4 of these
• Any night waking due to asthma?	Yes ☐ No ☐			
• SABA reliever for symptoms more than twice/week?*	Yes ☐ No ☐			
• Any activity limitation due to asthma?	Yes ☐ No ☐			

What is the difference between asthma control, risk and severity?

- The level of asthma control is the extent to which the features of asthma can be observed in the patient, or have been reduced or removed by treatment.
- Asthma control is assessed in two domains: *symptom control* and *future risk* of adverse outcomes. Poor symptom control reduces productivity and quality of life and increases the risk of exacerbations. Assessment of asthma control is useful in treatment titration. It is important to exclude suboptimal adherence and inhaler technique before up-titration of preventors.
- Asthma severity is assessed, after at least 2 - 3 months of adequate treatment, and defined as the level of treatment required to control symptoms and prevent exacerbations. Before labelling a patient as severe asthma, it is important to exclude causes of poor control. Note that patients with mild asthma (defined as patients who require only low dose ICS for control (GINA step 1 and 2)), can still have high risks of severe exacerbations if they are poorly controlled and rely only on SABA monotherapy.

How to assess a patient with asthma?

- Assess symptom control from the frequency of daytime and night-time asthma symptoms and SABA use, and from activity limitation. Symptom control tools include Asthma Control Test and Asthma Control Questionnaire.
- Assess the patient's future risk for exacerbations, even when symptom control is good. Risk factors for exacerbations that are independent of symptom control include a history of ≥ 1 exacerbation in the previous year, socioeconomic problems, poor adherence, incorrect inhaler technique, low lung function, smoking, and blood eosinophilia.
- Also assess risk factors for persistent airflow limitation and medication side-effects, treatment issues such as inhaler technique and adherence, and comorbidities, and ask the patient about their asthma goals
- Once the diagnosis of asthma has been made, lung function is most useful as an indicator of future risk. It should be recorded at diagnosis, 3–6 months after starting treatment, and periodically thereafter.
- Investigate further if there are few symptoms but impaired lung function, or frequent symptoms and good lung function.

Assessment of Asthma Control

Asthma Control Test (ACT)[®]

This is a 5-item, patient-administered questionnaire for assessing asthma control (*Figure 1*). It is a simple, objective, robust and validated method for monitoring control by doctors (and patients), which is being used internationally.

Asthma Control Test (ACT)

1. In the past 4 weeks, how much of the time did your asthma keep you from getting as much done at work, school or at home?

Score

All of the time 1 Most of the time 2 Some of the time 3 A little of the time 4 None of the time 5

2. During the past 4 weeks, how often have you had shortness of breath?

More than once a day 1 Once a day 2 3 to 6 times a week 3 Once or twice a week 4 Not at all 5

3. During the past 4 weeks, how often did your asthma symptoms (wheezing, coughing, shortness of breath, chest tightness or pain) wake you up at night or earlier than usual in the morning?

4 or more nights a week 1 2 or 3 nights a week 2 Once a week 3 Once or twice 4 Not at all 5

4. During the past 4 weeks, how often have you used your rescue inhaler or nebulizer medication (such as albuterol or salbutamol)?

3 or more times per day 1 1 or 2 times per day 2 2 or 3 times per week 3 Once a week or less 4 Not at all 5

5. How would you rate your asthma control during the past 4 weeks?

Not controlled at all 1 Poorly controlled 2 Somewhat controlled 3 Well controlled 4 Completely controlled 5

Patient Total Score

Based on a five-point scoring system, a maximum score of 25 will indicate 'total control' of asthma.

'Well controlled' asthma is defined as a score of 20 - 24, and a score of less than 20 will imply 'poor control'.

Figure 1: Asthma Control Test (ACT)[®] for adults and children aged 12 and above

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Lifestyle Modification and Patient Education

Important health education topics include:

- Self-monitoring and regular review
- Written action plan
- Modifiable risk factors and comorbidities, e.g. smoking, obesity, anxiety

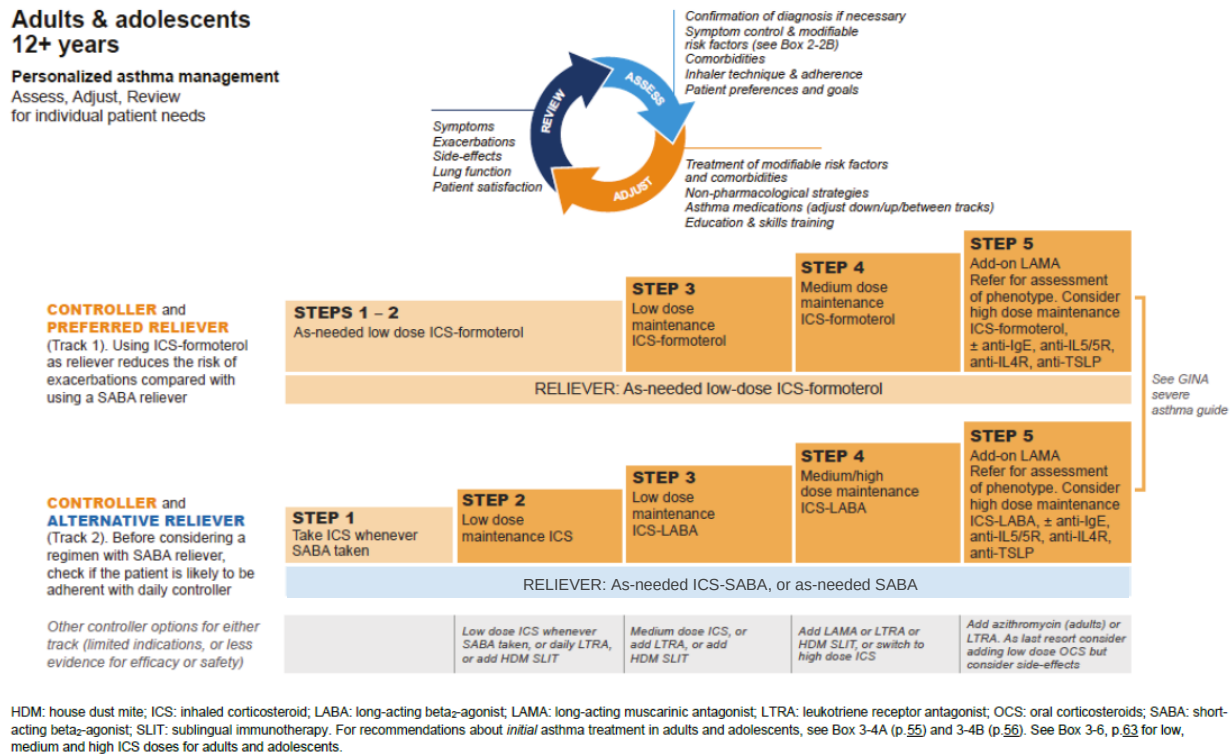
Non-Pharmacological Treatment

- Smoking cessation
- Physical activity for weight loss
- Avoidance of sensitizers where appropriate

Pharmacological Treatment

Adults & adolescents 12+ years

Personalized asthma management
Assess, Adjust, Review
for individual patient needs



Stepwise approach to asthma treatment

The patient's current treatment and level of control determine the selection of pharmacologic treatment. If asthma is not controlled on the current treatment, treatment should be stepped up until control is achieved. Control is usually maintained for at least 3 months before an attempt is made to step down the treatment, with the aim to establish the lowest step and dose of treatment that maintains control.

Figure 2 shows the steps 1 - 5 for achieving control. Each step represents treatment options in steps of increasing efficacy. In the management scheme described in Figure 2, the dose of daily asthma medication is adjusted according to the level of asthma control evaluated at each clinic visit.

Step 1:

- For safety, GINA and our local asthma guidelines no longer recommend SABA-only treatment for Step 1 (mild asthma)
- Regular low dose ICS, Low dose ICS taken whenever SABA is taken, or as-needed low dose ICS-formoterol
- Reserved for patients with infrequent symptoms (less than twice a month) of short duration, and with no risk factors for exacerbations

Step 2: Regular low dose ICS, or as-needed low dose ICS-formoterol

Step 3: Low dose ICS-LABA, or medium dose ICS. As needed low dose ICS-formoterol for patients prescribed maintenance and reliever therapy

Step 4: Medium dose ICS-LABA, or high dose ICS. As needed low dose ICS-formoterol for patients prescribed maintenance and reliever therapy

Step 5: Refer for specialist investigation and consideration of add-on treatment. **Management should be supervised directly by specialists.**

Low, Medium, High Daily Doses of Inhaled Corticosteroids

DPI: Dry powder inhaler; HFA: Hydrofluoroalkane propellant

Adults and Adolescents (12 years and older)

Drug	Low Daily Dose (mcg)	Medium Daily Dose (mcg)	High Daily Dose (mcg)
Beclomethasone Dipropionate (HFA)	100 - 200	> 200 - 400	> 400
Budesonide (DPI)	200 - 400	> 400 - 800	> 800
Fluticasone Propionate (DPI)	100 - 250	> 250 - 500	> 500
Fluticasone Propionate (HFA)	100 - 250	> 250 - 500	> 500
Fluticasone Furoate (DPI)	100	100	200

Note: LABAs should NOT be used without concomitant inhaled corticosteroids in asthma

REMEMBER TO:

- Provide guided self-management education (self-monitoring + written action plan + regular review).
- Treat modifiable risk factors and comorbidities, e.g. smoking, obesity, anxiety.
- Advise about non-pharmacological therapies and strategies, e.g. physical activity, weight loss, avoidance of sensitizers where appropriate.
- Consider stepping up if symptoms uncontrolled, exacerbations or risks, but check diagnosis, inhaler technique and adherence first.
- Consider referring to specialist if asthma is not well controlled on STEP 4 treatment.
- Consider stepping down if symptoms controlled for 3 months + low risk for exacerbations.
- Ceasing ICS is not advised.

For the list of medications for asthma management that are available in the NUP formulary, please refer to [NUP Intranet Asthma / COPD Medication Chart](#).

Role of Health Team Members

1. Family Physician

- Able to provide all aspects of primary medical care from screening, diagnosis and management of asthma, including health promotion and prevention / treatment of complications
- Collaborates with other health care providers in the health team to provide holistic and patient-centred care

2. Care Coordinator

- Perform general screening (i.e. fall risk, social economics, smoking & drinking history) and offer smoking cessation clinic
- Address care gaps under Health Maintenance Topics, such as vaccinations (influenza / pneumococcal)
- Perform GINA assessment of asthma control

3. Care Manager

- Evaluate understanding and provide education on asthma, good asthma control and lifestyle measures
- Assess and identify reasons for suboptimal / poor adherence to treatment plans and devise strategies to motivate the patients
- Provide education on preventer and reliever inhalers, inhaler technique and use of asthma action plan

4. Advanced Practice Nurse

- Manage patients with asthma within scope of practice
- Initiate and titrate medication according to stepwise approach to asthma treatment and within the scope of APN practice
- Initiate and educate patient on Written Asthma Action Plan
- Offer timely influenza and pneumococcal vaccinations
- Encourage smoking cessation

5. Dietitian

- Patient education on weight management

6. Psychologist

- Psychological and behavioural interventions to manage psychological stress, improve disease management and quality of life.
- Psychological assessment and intervention to manage co-occurring psychological problems (depression, anxiety disorders etc.)

7. Physiotherapist

- Assess and provide intervention for patients with MSK conditions to improve their mobility and ability to exercise
- Prescribe exercise and provide patient education on appropriate exercises for weight loss and to improve general fitness
- Patient education on bronchial hygiene and positions to ease shortness of breath

8. Pharmacist

- Assess and teach use of various inhalers and delivery devices
- Smoking cessation clinic
- Detect, prevent and address drug-drug/drug-disease interactions
- Perform medication reconciliation.

SPECIAL SITUATIONS

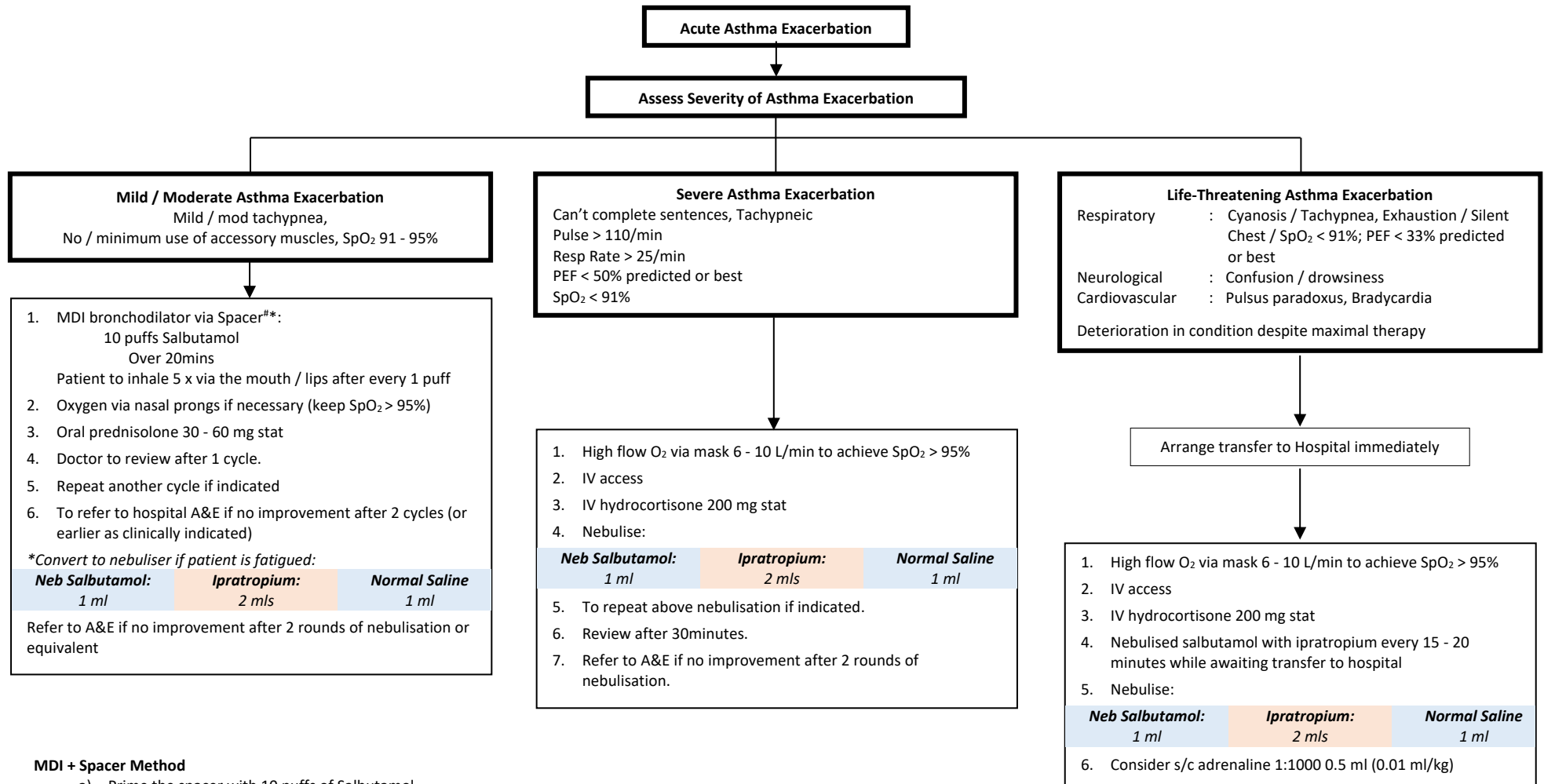
Asthma Exacerbations

1. How Severe is the Attack?

SEVERITY SYMPTOMS	MILD	MODERATE	SEVERE	RESPIRATORY ARREST IMMINENT
Breathless...	While walking Can lie down	While talking Prefer sitting	While at rest Hunched forward	
Talks in...	Sentences	Phrases	Words	
Alertness	May be agitated	Usually agitated	Usually agitated	Drowsy or confused
SEVERITY SIGNS	MILD	MODERATE	SEVERE	RESPIRATORY ARREST IMMINENT
Respiratory rate*	Increased	Increased	Often > 30/min	
Use of accessory muscles / suprasternal retractions	Usually not	Usually	Usually	Paradoxical thoraco– abdominal movement
Wheeze	Moderate, often only end expiratory	Loud	Usually loud; throughout inhalation and exhalation	Absence of wheeze
Pulse rate*	< 100/min	100 - 200/min	> 120/min	Bradycardia
FUNCTIONAL ASSESSMENT				
PEF	> 80%	Approx. 60 - 80%	< 60% predicted or personal best in severe asthma exacerbation	
Pulse Oximetry SpO ₂ (on air)	> 95%	91 - 95%	< 90%	

* The presence of several parameters, but not necessarily all, indicate the general classification of the attacks. Note also any prior usage of bronchodilator as this may alter the presenting clinical pictures.

2. Management of Acute Exacerbation in Adults



MDI + Spacer Method

- Prime the spacer with 10 puffs of Salbutamol
- Load the spacer with 1 puff each time, patient to inhale 5 times (tidal breaths) after every 1 puff (if patient can cooperate, deep breaths with breath holding recommended).
- Oxygen can be administered concurrently via nasal prongs if required – maintain SpO₂ > 95%
- Nurse to administer puffs, ensure inhalation via the mouth / lips
- Patient can self-administer bronchodilator treatment with supervision by medical staff

Repeat Clinical Assessment after Bronchodilator Treatment
Symptoms, physical examination, O₂ saturation, PEF, other tests as needed

Good Response

1. Response sustained 60 minutes after last treatment
2. Physical examination is normal
3. PEF > 70% predicted
4. No stress
5. O₂ saturation > 90%

Action:

1. Discharged
2. Continue treatment with inhaled β_2 -agonist.
3. Consider course of prednisolone 30 mg om for 5 - 7 days in most cases.
4. Initiate or continue inhaled glucocorticosteroids.
5. Reinforce patient education, action plan and close follow-up.

Incomplete Response

- History of high-risk patient
- Physical examination:
1. Mild to moderate symptoms
 2. PEF > 50% - 70%
 3. O₂ saturation not improving

Action:

1. Refer to Hospital A&E
2. O₂ via mask 6 - 10 L/min to achieve SpO₂ > 95%
3. Nebulised salbutamol with ipratropium every 15- 20 minutes while awaiting transfer to hospital
4. i/v Hydrocortisone 200 mg stat if not already administered earlier

Poor Response

- History of high-risk patient
- Physical examination:
1. Symptoms severe, drowsiness, confusion
 2. PEF < 30%
 3. O₂ saturation < 90%

Action:

1. **ARRANGE URGENT TRANSFER TO HOSPITAL via ambulance immediately**
2. High flow O₂ via mask 6 - 10 L/min to achieve SpO₂ > 95%
3. Nebulised salbutamol with ipratropium every 15 - 20 minutes while awaiting transfer to hospital
4. i/v Hydrocortisone 200 mg if not already administered earlier
5. Consider s/c adrenaline 1:1000 0.5 ml (0.01 ml/kg)
6. Possible intubation & mechanical ventilation

Asthma: Emergency Drug List

1. Salbutamol Inhaler / neb
2. Prednisolone tab
3. Hydrocortisone IV
4. Ipratropium bromide nebuliser
5. Adrenaline IM

REFERRALS

When to Refer to Respiratory Specialist

Presence of Risk Factors for Death from Asthma

- Prior intubation and mechanical ventilation for asthma
- Hospitalisation or emergency care visit for asthma in the past year
- Current use of systemic corticosteroids or recent withdrawal from systemic corticosteroids
- Not currently using inhaled corticosteroids
- Use of > 1 canister of inhaled short-acting β 2-agonist within 1-2 months
- History of psychiatric disease or psychosocial problems

Patients at high risk of dying with asthma require special attention, monitoring and care, particularly intensive education, including advice to seek medical care early during an exacerbation.

Acute Asthma - Severe or Frequent Exacerbations.

1. A life-threatening asthma exacerbation.
2. Frequent exacerbations:
 - a) acute exacerbations 2 - 3 times a year, or
 - b) more than once every six months, despite compliance with medications and good inhaler (or inhaler and device) technique
3. Need for continuous oral corticosteroid therapy or not well-controlled on Step 4 therapy

Chronic Asthma - Difficult or Poor Control

1. Failing goals of therapy after 3 to 6 months of treatment.
2. Uncontrolled Asthma.

3. Steroid use:
 - a) Continuous oral corticosteroid therapy, or
 - b) Require more than two bursts of oral corticosteroids in 1 year, or
 - c) High-dose inhaled corticosteroids.

Diagnosis

1. Atypical signs and symptoms.
2. Other conditions complicate asthma or its diagnosis, e.g. heart failure, COPD, unsure of diagnosis.
3. Additional diagnostic testing is indicated.
4. Suspicion of occupational asthma.

RECOMMENDED CARE COMPONENTS FOR ASTHMA

Recommended Care Components	Minimum Frequency	Remarks
Asthma Control Assessment [^] [Global Initiative for Asthma (GINA) Score, Asthma Control Test*]	At least twice a year	
Smoking Assessment [^]	Annually for smokers; Once-off for non-smokers, unless there is a change in smoking habit	Assessment on smoking habits (estimated sticks/day; zero for non- or ex-smoker) and provide smoking cessation counselling
Written Asthma Action Plan	Upon diagnosis, recommended annually	
Spirometry	Recommended at or soon after diagnosis, or when clinically indicated	
Influenza Vaccination and Pneumococcal Vaccination	As recommended under the National Adult Immunisation Schedule	

[^] *These are reportable clinical indicators*

* *Asthma Control Test is not used for asthma control assessment in NUP*

ABBREVIATIONS

<	Less than
≤	Less than or equal to
>	More than
≥	More than or equal to
ABPA	Allergic Bronchopulmonary Aspergillosis
BDP	Beclomethasone Dipropionate
BUD	Budesonide
Ca	Carcinoma
COPD	Chronic Obstructive Pulmonary Disease
CT	Computed Tomography
CXR	Chest X-Ray
EGPA	Eosinophilic Granulomatosis with Polyangiitis
ETT	Endotracheal tube
FEV1	Forced expiratory volume in 1 second
GERD	Gastroesophageal Reflux Disease
GINA	Global Initiative for Asthma
i/v	Intravenous
ICS	Inhaled Corticosteroids
LABA	Long-Acting Beta-Agonists
LOW	Loss of weight
MDI	Metered Dose Inhaler
mths	Month(s)
O ₂	Oxygen
OGD	Oesophago-Gastro-Duodenoscopy
PEF	Peak expiratory flow
s/c	Subcutaneous
SABA	Short-Acting Beta-Agonists
TCU	To-see-you (i.e. review appointment)
Yr	Year(s)

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